

BDC v 2.0 UI MANUAL

Embedded Keypad + Display System

Project Name:

BDC v 2.0

Module:

User Interface (UI)

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UI MANUAL – EMBEDDED KEYPAD + DISPLAY SYSTEM

1. Overview

The User Interface (UI) is a state-based menu system designed for embedded applications.

It is driven by:

- A keypad (Up, Down, OK, Back, *, #, digits)
- A small OLED display using U8g2 library
- Multiple screens controlled using screen states

Each screen consists of:

- A draw function (UI rendering)
- A handle function (key processing)

2. Keypad Function Map

The following keys are used for system navigation and interaction.

Global Keys

Key	Function
UP	Move Up / Previous item
DOWN	Move Down / Next item
OK	Select / OK / Enter
BACK	Back / Exit
*	Context action (Route / Dot / Backspace)
#	Trip toggle (Start / Stop)
0–9	Numeric input

3. Screen List

The following screens define the system interface and user interaction.

3.1 Home Screen

Displayed Information

- Route number
- Route name
- Logged user
- Server ID

- Device status (LED / CAM / HDD / GPS / ETH)
- Wi-Fi signal strength

Key Actions

Key	Action
OK	Open Main Menu
*	Open Route Selection
#	Start / End Trip

Trip Logic

- First press of # → Trip **STARTED**
- Second press of # → Trip **ENDED**



Figure 1: Home Screen

3.2 Main Menu Screen

The Main Menu provides access to all primary system functions.

Options

- Login
- OCC Messages
- Route Select
- Internal Messages
- Manual
- Schedule
- Settings (Admin only)

Key Actions

Key	Action
UP / DOWN	Navigate menu
OK	Enter selected menu
BACK	Return to Home Screen

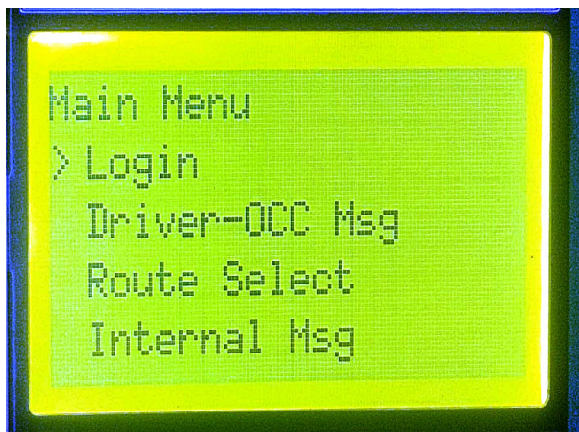


Figure 2: Main Menu – 1



Figure 3: Main Menu - 2

3.3 Login Menu Screen

The Login Menu allows users to access the system based on their role.

Options

- Driver Login
- Admin Login
- Maintenance Login
- Logout

Flow

- Select login type
- Enter user ID
- Enter password
- Validate credentials
- Set user role
- Upon successful login, the system redirects to the Home Screen

Key Actions

Key	Action
UP / DOWN	Navigate
OK	Select
BACK	Return to Previous Screen

Logout Behavior

- Sends logout message (**D_LOGOUT / A_LOGOUT / M_LOGOUT**)
- Resets user session



Figure 4: Login Menu Screen

3.4 Login ID Screen

This screen allows the user to enter their login ID.

Input

- Numeric ID only (maximum 5 digits)

Key Actions

Key	Action
0–9	Enter digits
OK	Proceed to Password Screen
BACK	Return to Login Menu

3.5 Login Password Screen

This screen allows secure password entry with masked display.

Input

- Numeric password (maximum 5 digits)
- Display masked (****)

Key Actions

Key	Action
0–9	Enter password
OK	Validate login
BACK	Return to Previous Screen

3.6 Route Selection Screen

This screen allows the user to select and update the route number.

Input

- Enter route number

Key Actions

Key	Action
0–9	Enter route number
OK	Confirm route
BACK	Cancel and return to previous screen

On Confirmation

- Sends RS485 message
- Sends IR encoded digits
- Posts route update to system queue
- Displays popup message **“Route Changed”**

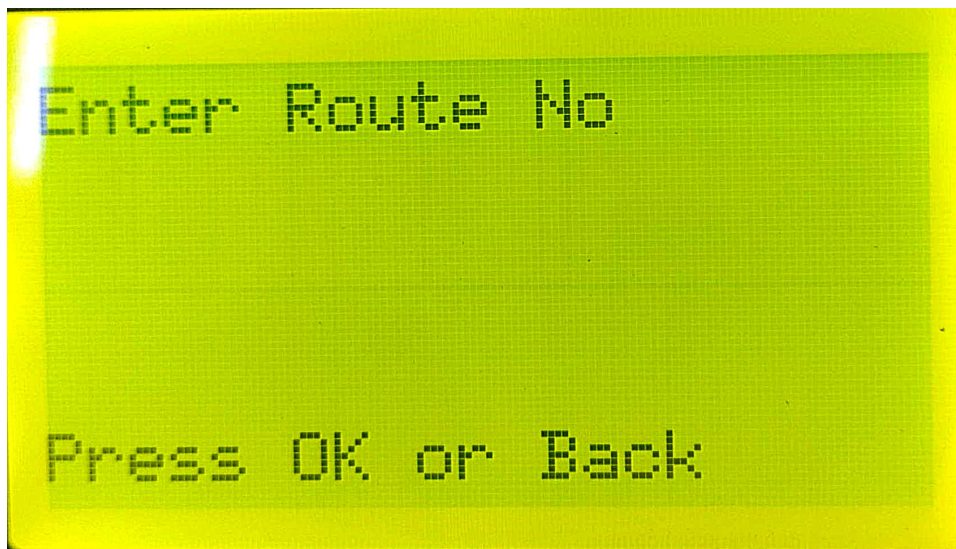


Figure 5: Route Selection Screen

3.7 Settings Screen (Admin Only)

This screen allows configuration of system parameters and administrative settings.

Features

- Server configuration
- Port configuration
- Active server switch
- Wi-Fi SSID display
- Device ID editing
- Firmware version

Key Actions

Key	Action
UP / DOWN	Move through settings
OK	Edit or toggle selected option
BACK	Exit to previous screen

Editable Fields

- Server 1 IP
- Server 2 IP
- Port 1 / Port 2
- Device ID (15 digits)

Validation Rules

- IP address must be valid IPv4 format
- Port range: **1–65535**
- Device ID must be exactly **15 digits**

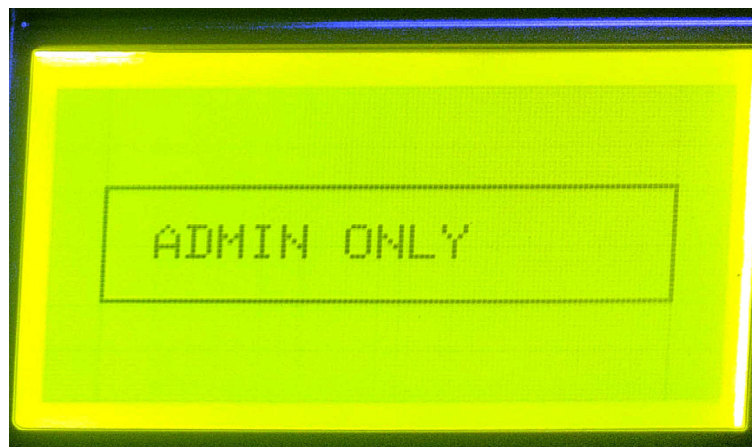


Figure 6: Setting Screen

3.8 Settings Edit Mode

This mode allows the user to modify configuration values.

Key Actions

Key	Action
0–9	Enter numeric values
*	Insert “.” (for IP) / Backspace (for Device ID)
OK	Save changes
BACK	Cancel editing

Save Behavior

- Displays popup message “**SAVED**”
- Stores configuration in flash memory

3.9 OCC Message Screen

This screen allows the user to send predefined operational messages.

Key Actions

Key	Action
UP / DOWN	Navigate messages
OK	Send selected message
BACK	Return to Previous Screen



Figure 7: OCC MSG Screen - 1

Figure 8: OCC MSG Screen - 2

Figure 9: OCC MSG Screen - 3

3.10 Internal Message Screen

This screen allows the user to view and send internal system messages.

Key Actions

Key	Action
UP / DOWN	Navigate messages
OK	Select message
BACK	Return to Previous Screen



Figure 10: Internal Message Screen

3.11 Schedule Screen

This screen displays the current schedule information.

Displayed Information

- Current schedule item
- Index (e.g., 2/10)

Behavior

- Displays one schedule item at a time
- Index updates dynamically (current/total)

Key Actions

Key	Action
UP	Previous schedule
DOWN	Next schedule
OK	Exit to previous screen

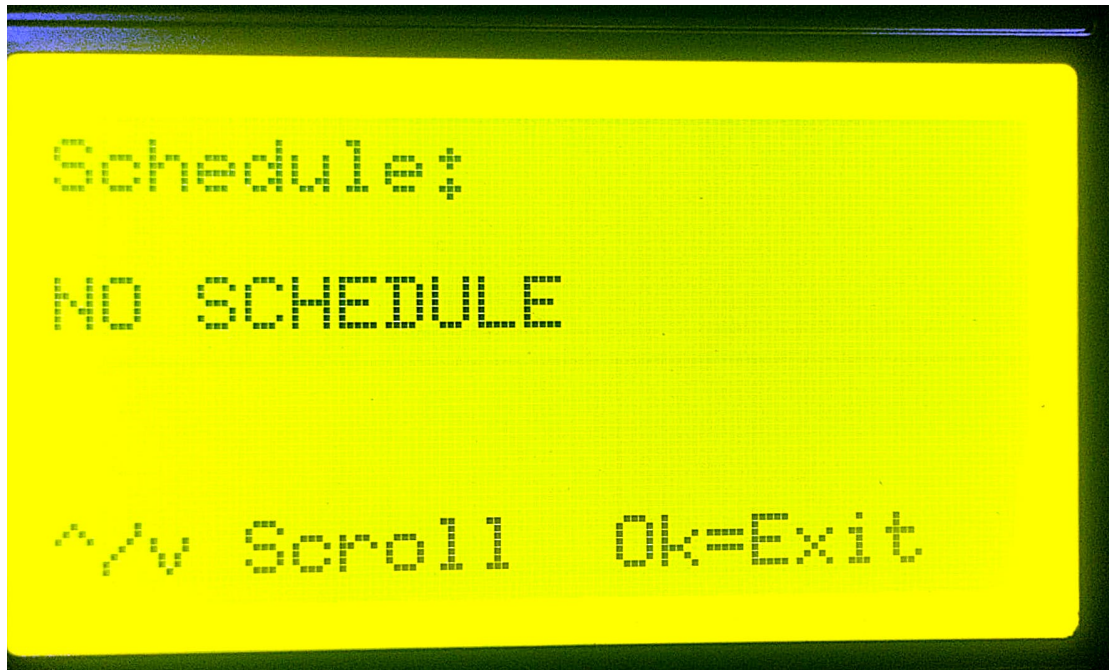


Figure 11: Schedule Screen

3.12 Manual Screen

This screen displays a static help page for user reference.

Displayed Information

- Static help content

Behavior

- Read-only screen
- No navigation except exit

Key Actions

Key	Action
BACK	Return to Main Menu

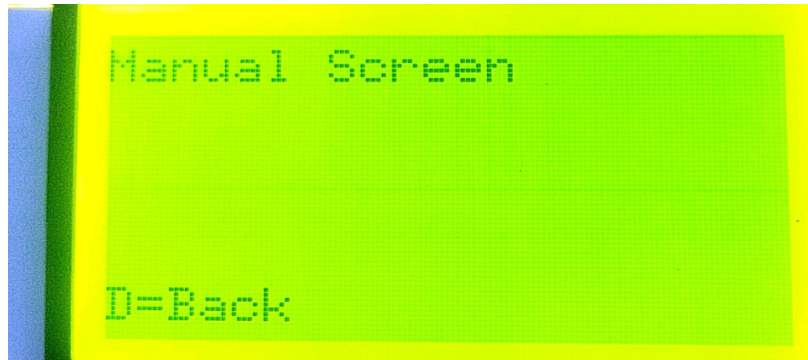


Figure 12: Manual Screen

3.13 Popup System

This system displays temporary messages for user feedback.

Behavior

- Displays a centered message box
- Automatically closes after a predefined duration
- Optional beep alert every 400 ms

System Rules

- Popup has highest display priority
- Blocks background screen interaction
- Automatically restores previous screen after timeout

Common Popup Messages

- "Route Changed"
- "Trip STARTED"
- "Trip ENDED"
- "INVALID LOGIN"
- "SAVED"

3.14 Wi-Fi Indicator

This indicator shows the signal strength of the Wi-Fi connection.

Behavior

- RSSI converted to percentage
- Displayed as 0–5 bar icon
- Updated periodically

3.15 Audio Feedback

This system provides sound feedback for user actions.

Events and Sounds

Event	Sound
MOVE	Beep Move
SELECT	Beep Select
BACK	Beep Back
ALERT	Periodic Beep (popup)

3.16 Role Restrictions

This defines access control based on user roles.

Rules

- Settings screen is accessible to **Admin only**
- Non-admin access displays popup: **“ADMIN ONLY”**

If unauthorized:

- Popup: **“ADMIN ONLY”**
- Block access

4. System Behavior Summary

UI Loop

- Read keypad input
- Route to screen handler
- Draw UI if redraw = true
- Handle popup and Wi-Fi updates
- Delay 20 ms (RTOS task loop)

System Characteristics

- Event-driven architecture
- State-machine UI
- Non-blocking design
- Background tasks supported

5. Navigation Flow (FSM)

